

- information in the form of a concentration gradient along an embryonic axis.
- morphogenesis** (mōr'-fō-jen'-uh-sis) The development of the form of an organism and its structures.
- morphological species concept** Definition of a species in terms of measurable anatomical criteria.
- moss** A small, herbaceous, nonvascular plant that is a member of the phylum Bryophyta.
- motor neuron** A nerve cell that transmits signals from the brain or spinal cord to muscles or glands.
- motor protein** A protein that interacts with cytoskeletal elements and other cell components, producing movement of the whole cell or parts of the cell.
- motor system** An efferent branch of the vertebrate peripheral nervous system composed of motor neurons that carry signals to skeletal muscles in response to external stimuli.
- motor unit** A single motor neuron and all the muscle fibers it controls.
- movement corridor** A series of small clumps or a narrow strip of quality habitat (usable by organisms) that connects otherwise isolated patches of quality habitat.
- MPF** Maturation-promoting factor (or M-phase-promoting factor); a protein complex required for a cell to progress from late interphase to mitosis. The active form consists of cyclin and a protein kinase.
- mucoromycete** A member of the fungal phylum Mucoromycota, characterized by the formation of a sturdy structure called a zygosporangium during sexual reproduction.
- mucus** A viscous and slippery mixture of glycoproteins, cells, salts, and water that moistens and protects the membranes lining body cavities that open to the exterior.
- Müllerian mimicry** (myū-lār'-ē-un mim'-uh-krē) Reciprocal mimicry by two unpalatable species.
- multifactorial** Referring to a phenotypic character that is influenced by multiple genes and environmental factors.
- multigene family** A collection of genes with similar or identical sequences, presumably of common origin.
- multiple fruit** A fruit derived from an entire inflorescence.
- multiplication rule** A rule of probability stating that the probability of two or more independent events occurring together can be determined by multiplying their individual probabilities.
- muscle tissue** Tissue consisting of long muscle cells that can contract, either on its own or when stimulated by nerve impulses.
- mutagen** (myū'-tuh-jen) A chemical or physical agent that interacts with DNA and can cause a mutation.
- mutation** (myū-tā'-shun) A change in the nucleotide sequence of an organism's DNA or in the DNA or RNA of a virus.
- mutualism** (myū'-chū-ul-izm) A +/+ ecological interaction that benefits individuals of both interacting species.
- mycelium** (mī-sē'-lē-um) (plural, **mycelia**) The densely branched network of hyphae in a fungus.
- mycorrhiza** (plural, **mycorrhizae**) (mī'-kō-rī'-zuh, mī'-kō-rī'-zē) A mutualistic association of plant roots and fungus.
- mycosis** (mī-kō'-sis) General term for a fungal infection.
- myelin sheath** (mī'-uh-lin) Wrapped around the axon of a neuron, an insulating coat of cell membranes from Schwann cells or oligodendrocytes. It is interrupted by nodes of Ranvier, where action potentials are generated.
- myofibril** (mī'-ō-fī'-bril) A longitudinal bundle in a muscle cell (fiber) that contains thin filaments of actin and regulatory proteins and thick filaments of myosin.
- myoglobin** (mī'-uh-glō'-bin) An oxygen-storing, pigmented protein in muscle cells.
- myosin** (mī'-uh-sin) A type of motor protein that associates into filaments that interact with actin filaments to cause cell contraction.
- myriapod** (mir'-ē-uh-pod') A terrestrial arthropod with many body segments and one or two pairs of legs per segment. Millipedes and centipedes are the two major groups of living myriapods.
- NAD⁺** The oxidized form of nicotinamide adenine dinucleotide, a coenzyme that can accept electrons, becoming NADH. NADH temporarily stores electrons during cellular respiration.
- NADH** The reduced form of nicotinamide adenine dinucleotide that temporarily stores electrons during cellular respiration. NADH acts as an electron donor to the electron transport chain.
- NADP⁺** The oxidized form of nicotinamide adenine dinucleotide phosphate, an electron carrier that can accept electrons, becoming NADPH. NADPH temporarily stores energized electrons produced during the light reactions.
- NADPH** The reduced form of nicotinamide adenine dinucleotide phosphate; temporarily stores energized electrons produced during the light reactions. NADPH acts as "reducing power" that can be passed along to an electron acceptor, reducing it.
- natural killer cell** A type of white blood cell that can kill tumor cells and virus-infected cells as part of innate immunity.
- natural selection** A process in which individuals that have certain inherited traits tend to survive and reproduce at higher rates than other individuals because of those traits.
- negative feedback** A form of regulation in which accumulation of an end product of a process slows the process; in physiology, a primary mechanism of homeostasis, whereby a change in a variable triggers a response that counteracts the initial change.
- negative pressure breathing** A breathing system in which air is pulled into the lungs.
- nematocyst** (nem'-uh-tuh-sist') In a cnidocyte of a cnidarian, a capsule-like organelle containing a coiled thread that when discharged can penetrate the body wall of the prey.
- nephron** (nēf'-ron) The tubular excretory unit of the vertebrate kidney.
- neritic zone** The shallow region of the ocean overlying the continental shelf.
- nerve** A fiber composed primarily of the bundled axons of neurons.
- nervous system** In animals, the fast-acting internal system of communication involving sensory receptors, networks of nerve cells, and connections to muscles and glands that respond to nerve signals; functions in concert with the endocrine system to effect internal regulation and maintain homeostasis.
- nervous tissue** Tissue made up of neurons and supportive cells.
- net ecosystem production (NEP)** The gross primary production of an ecosystem minus the energy used by all autotrophs and heterotrophs for respiration.
- net primary production (NPP)** The gross primary production of an ecosystem minus the energy used by the producers for respiration.
- neural crest** In vertebrates, a region located along the sides of the neural tube where it pinches off from the ectoderm. Neural crest cells migrate to various parts of the embryo and form pigment cells in the skin and parts of the skull, teeth, adrenal glands, and peripheral nervous system.
- neural tube** A tube of infolded ectodermal cells that runs along the anterior-posterior axis of a vertebrate, just dorsal to the notochord. It will give rise to the central nervous system.
- neurohormone** A molecule that is secreted by a neuron, travels in body fluids, and acts on specific target cells, changing their functioning.
- neuron** (nyūr'-on) A nerve cell; the fundamental unit of the nervous system, having structure and properties that allow it to conduct signals by taking advantage of the electrical charge across its plasma membrane.
- neuronal plasticity** The capacity of a nervous system to change with experience.
- neuropeptide** A relatively short chain of amino acids that serves as a neurotransmitter.
- neurotransmitter** A molecule that is released from the synaptic terminal of a neuron at a chemical synapse, diffuses across the synaptic cleft, and binds to the postsynaptic cell, triggering a response.
- neutral variation** Genetic variation that does not provide a selective advantage or disadvantage.
- neutron** A subatomic particle having no electrical charge (electrically neutral), with a mass of about 1.7×10^{-24} g, found in the nucleus of an atom.
- neutrophil** The most abundant type of white blood cell. Neutrophils are phagocytic and tend to self-destruct as they destroy foreign invaders, limiting their life span to a few days.
- nitric oxide (NO)** A gas produced by many types of cells that functions as a local regulator and as a neurotransmitter.
- nitrogen cycle** The natural process by which nitrogen, either from the atmosphere or from decomposed organic material, is converted by soil bacteria to compounds assimilated by plants. This incorporated nitrogen is then taken in by other organisms and subsequently released, acted on by bacteria, and made available again to the nonliving environment.
- nitrogen fixation** The conversion of atmospheric nitrogen (N_2) to ammonia (NH_3). Biological nitrogen fixation is carried out by certain prokaryotes, some of which have mutualistic relationships with plants.